# MTH405

# QUIZ 1 UPDATED 22 MAY 2023

1. The order of the group {1, ω, ω^2} is:

a) 4

b) 3

c) 2

d) 5

Correct option: c) 2

2. (R, +) is not a subgroup of (C, +). Is this statement true or false?

a) False

b) True

Correct option: b) True

3. (Z, +) is a subgroup of (Q, +). Is this statement true or false?

a) False

b) True

Correct option: b) True

4. Let the order of a group be 27. Which of the following CANNOT be the order of a subgroup of G?

a) 1

b) 3

c) 9

d) 7

Correct option: d) 7

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5. Which of the following is NOT a rational number?

a) 3

b) 22/7

c) π

d) 0

Correct option: c) π

6. If a, b are elements of group G, then (a^(-1)b)^(-1) is equal to:

a) a^(-1)b^(-1)

b) a^(-1)b

c) ba^(-1)

d) b^(-1)a

Correct option: d) b^(-1)a

7. If G = {1, -1, i, -i} is a group under multiplication, then the inverse of i is:

a) i

b) -i

c) -1

d) 1

Correct option: b) -i

8. If X has 5 elements, then the total number of permutations on X is:

a) 5

b) 5+1

c) 5!

d) 5^2

Correct option: c) 5!

9. The group ({1, -1}, *) is a subgroup of the group (Z, +). Is this statement true or false?

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a) True

b) False

Correct option: b) False

10. Is every cyclic group an abelian group?

a) True

b) False

Correct option: a) True

